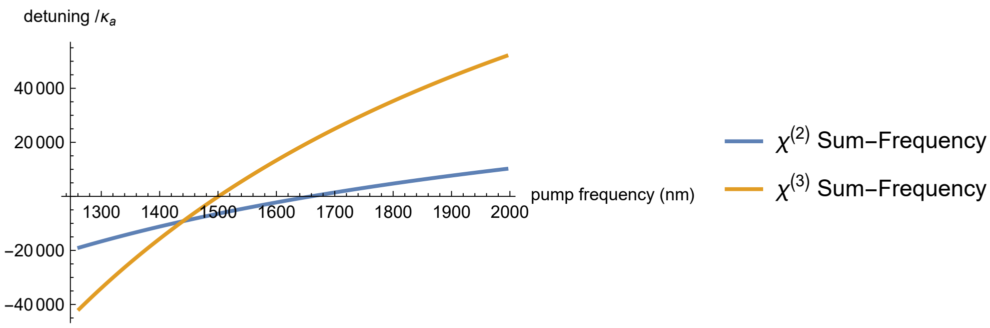
```
In[180]:= (* All time and rate is normalized by κ *)
              全部
          (* Number of span modes *)
              数
          dir = NotebookDirectory[];
                当前笔记本的目录

          SetDirectory[dir];
          设置目录

          hbar = 1.054 * 10^-34;
          Nspan = 60;
          Ntot = 2 * Nspan + 1; (*Total number of modes*)
                                 总计
          (*Dispersion*)
          d1a = (4.5665) * 10^12; (*1550nm*)
          d1b = 4.4189 * 10^12; (*775nm*)
          d1c = (4.25) * 10^12; (*515nm*)

          d2a = 2.0 * 8.84 * 10^8; (*1550nm*)
          d2b = 2.0 * 4.31 * 10^8; (*775nm*)
          d2c = 2.0 * 2.31 * 10^8; (*515nm*)


          (* Mode frequency and dissipation rate *)
          ω0a = 2 π * ((3 * 10^8) / (1550. * 10^-9));
          nboff = -40;
          ncoff = 25;
          (*Ω0=2ω0+0.0(Random[]-0.5)*(d1-D1)+2(noff*d1+noff^2*d2/2)-(2noff*D1+4noff^2*D2/2);*)
          (*Here is also need to be checked maybe not good*)
            当前位置
          ω0b = 2 ω0a + (nboff * d1a + nboff^2 * d2a / 2) - (nboff * d1b + nboff^2 * d2b / 2);
          ω0c = 3 ω0a + (ncoff * d1a + ncoff^2 * d2a / 2) - (ncoff * d1c + ncoff^2 * d2c / 2);
          ωa[j_] := ω0a + d1a * j + (d2a / 2.) * j^2;
          ωb[j_] := ω0b + d1b * j + (d2b / 2.) * j^2;
          ωc[j_] := ω0c + d1c * j + (d2c / 2.) * j^2;
          κa = ω0a / (2 * 500. * 10^3);
          κa1 = κa0 = κa / 2.;
          κb = ω0b / (2 * 70. * 10^3);
          κb1 = κb0 = κb / 2.;
          κc = ω0c / (2 * 70. * 10^3);
          κc1 = κc0 = κc / 2.;
          ωr = 2 π * 18.08 * 10^12;
          (*what is this r mode here?*) κr = 2 π * 60 * 10^9;
```

In[199]:=
```
(*Plot the phase matching*)
 绘图

Print[ListLinePlot[{Table[{(2 π * 3. * 10^8)/(ωa[z]) * 10^9, (ωb[2 z] - 2 ωa[z])/(κa)}, {z, -Nspan, Nspan}],
打印  绘制点集的线条  表格
    Table[{(2 π * 3. * 10^8)/(ωa[z]) * 10^9, (ωc[3 z] - 3 ωa[z])/(κa)}, {z, -Nspan, Nspan}]}, PlotStyle →
    表格                                                                                    绘图样式
    Directive[Thick], PlotLegends → {"χ^(2) Sum-Frequency", "χ^(3) Sum-Frequency"},
    指令      粗       绘图的图例          求和                  求和
   AxesLabel → {"pump frequency (nm)", "detuning /κa"}]];
   坐标轴标签
```

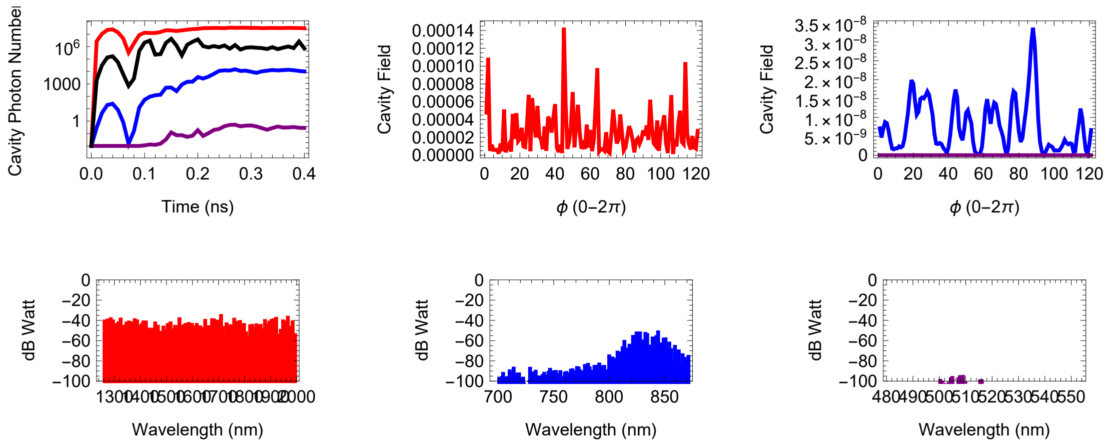


In[200]:=
```
(* nonlinear coupling terms *)
g2aab = 2 π * 0.08 * 1.0 * 10^6 * 1.00;
g2abc = 2 π * 0.08 * 1.0 * 10^6 * 0.10;
g3aa = -2 π * 2.8 * 1.00;
g3bb = -2 π * 14.9 * 1.00;
g3ab = -2 π * 17.1 * 1.00;
g3ac = -2 π * 0.1 * 1.00;
gR = 2 π * 0.0003 * 10^6;
(* pump to 1550 *)
p = 0; (* pump mode number *)
Pin = 1.0 * 2.9; (*Watt*) (*External Driving*)
ϵ = Sqrt[(2 κa1 * Pin)/(hbar * ω0a)]; (*normalized driving amplitude*)
```

In[210]:=
```
(*Initial cavity field*)
EA = EA0 = Table[{10^(-10-10*Random[])}, {i, 1, Ntot, 1}];
           表格
EB = EC = ER = Table[{0.0}, {i, 1, Ntot, 1}]; (*why noise here*)
               表格
(* Rearrange the frequency for FFT *)
liωa = liωb = liωc = {};
For[i = 1, i ≤ Ntot, i += 1,
For循环
  j = i - 1;
  If[i ≥ Nspan + 2, j = i - 1 - Ntot];
  如果
  AppendTo[liωa, {ωa[j]}];
  附加
  AppendTo[liωb, {ωb[j]}];
  附加
  AppendTo[liωc, {ωc[j]}]
  附加
 ];
```

In[214]:=
```
(*Normalize Photon Number by*)
  正规化               数
Anorm = 1.0 * 10^5;
(* Normalize time by *)
   正规化
κnorm = κa;
CombA = Table[{(2 π * 3. * 10^8)/(liωa[[i, 1]]) * 10^9,
        表格
   10 * Log[10, hbar * ω0a * 2 κa1 * Abs[EA[[i, 1]]]^2 * Abs[Anorm]^2 + 10^-199]}, {i, 1, Ntot}];
        对数
CombB = Table[{(2 π * 3. * 10^8)/(liωb[[i, 1]]) * 10^9,
        表格
   10 * Log[10, hbar * ω0b * 2 κb1 * Abs[EB[[i, 1]]]^2 * Abs[Anorm]^2 + 10^-199]}, {i, 1, Ntot}];
        对数
CombC = Table[{(2 π * 3. * 10^8)/(liωc[[i, 1]]) * 10^9,
        表格
   10 * Log[10, hbar * ω0c * 2 κb1 * Abs[EC[[i, 1]]]^2 * Abs[Anorm]^2 + 10^-199]}, {i, 1, Ntot}];
        对数
CombAav = CombBav = CombCav = Table[{0.0}, {i, 1, Ntot}];
                              表格
```

In[220]:=
```
liNpA = liNpB = liNpC = liNpR = {};
Print[Dynamic[GraphicsGrid[
|打印  |动态     |图形网格
  {{ListLogPlot[{liNpA, liNpB, liNpC, liNpR}, PlotRange → All, Joined → True,
    |点集的对数图                                   |绘制范围    |全部 |连接点   |真
     Axes → False, Frame → True, FrameLabel → {"Time (ns)", "Cavity Photon Number"},
     |坐标轴 |假     |边框    |真    |边框标签                                      |数
     PlotStyle → {Directive[Thick, Red], Directive[Thick, Blue],
     |绘图样式    |指令      |粗    |红色 |指令      |粗    |蓝色
       Directive[Thick, Purple], Directive[Thick, Black]}, ImageSize → 300],
       |指令      |粗    |紫色    |指令      |粗    |黑色    |图像尺寸
    ListLinePlot[{lica[[1]]}, PlotRange → All, Joined → True, Axes → False,
    |绘制点集的线条          |绘制范围   |全部 |连接点  |真     |坐标轴 |假
     Frame → True, FrameLabel → {"ϕ (0-2π)", "Cavity Field"},
     |边框    |真     |边框标签
     PlotStyle → {Directive[Thick, Red], Directive[Thick, Blue],
     |绘图样式    |指令      |粗    |红色 |指令      |粗    |蓝色
       Directive[Thick, Purple], Directive[Thick, Black]}, ImageSize → 300],
       |指令      |粗    |紫色    |指令      |粗    |黑色    |图像尺寸
    ListLinePlot[{licb[[1]], licc[[1]]}, PlotRange → All, Joined → True,
    |绘制点集的线条                     |绘制范围   |全部 |连接点   |真
     Axes → False, Frame → True, FrameLabel → {"ϕ (0-2π)", "Cavity Field"},
     |坐标轴 |假     |边框    |真    |边框标签
     PlotStyle → {Directive[Thick, Blue], Directive[Thick, Purple],
     |绘图样式    |指令      |粗    |蓝色  |指令      |粗    |紫色
       Directive[Thick, Black]}, ImageSize → 300]},
       |指令      |粗    |黑色    |图像尺寸
   {ListPlot[CombA, PlotStyle → Red, Filling → -100, FillingStyle → Directive[
    |绘制点集      |绘图样式    |红色 |填补          |填充样式       |指令
       Thick, Red], ImageSize → 300, AspectRatio → 1/2, PlotRange → {All, {-100, 0}},
       |粗    |红色  |图像尺寸       |宽高比           |绘制范围    |全部
     Frame → True, FrameLabel → {"Wavelength (nm)", "dB Watt"}, Axes → False],
     |边框    |真    |边框标签                                    |坐标轴 |假
    ListPlot[CombB, PlotStyle → Blue, Filling → -200,
    |绘制点集      |绘图样式    |蓝色  |填补
     FillingStyle → Directive[Thick, Blue], ImageSize → 300,
     |填充样式       |指令      |粗    |蓝色   |图像尺寸
     AspectRatio → 1/2, PlotRange → {All, {-100, 0}}, Frame → True,
     |宽高比           |绘制范围    |全部              |边框    |真
     FrameLabel → {"Wavelength (nm)", "dB Watt"}, Axes → False], ListPlot[CombC,
     |边框标签                                    |坐标轴 |假       |绘制点集
     PlotStyle → Purple, Filling → -200, FillingStyle → Directive[Thick, Purple],
     |绘图样式    |紫色    |填补          |填充样式       |指令      |粗    |紫色
     ImageSize → 300, AspectRatio → 1/2, PlotRange → {All, {-100, 0}}, Frame → True,
     |图像尺寸         |宽高比           |绘制范围    |全部              |边框    |真
     FrameLabel → {"Wavelength (nm)", "dB Watt"}, Axes → False]}}]]];
     |边框标签                                    |坐标轴 |假
```



In[222]:=
```
δ0 = -15.0;
dδ = 0.;
(*δ0+=dδ*dt/T; and what is this?*)
δ = δ0 * κa;
Δa[j_] := d2a/2 * (j^2 - p^2) - δ;
(*Δa[j_]:=ω0a+d1a*j+d2a/2*j^2-1(ω0a+p*d1a+d2a/2*p^2+δ)-(j-p)d1a;*)
Δb[j_] := ω0b + d1b * j + d2b/2 * j^2 - 2 (ω0a + p * d1a + d2a/2 * p^2 + δ) - (j - 2 p) d1a;
Δc[j_] := ω0c + d1c * j + d2c/2 * j^2 - 3 (ω0a + p * d1a + d2a/2 * p^2 + δ) - (j - 3 p) d1a;
Δr[j_] := Sign[ωr - j * d1a] * Min[Abs[ωr - j * d1a], 30 κr];
          正负符号                …  绝对值
```

In[229]:=
```
(* Matrix of pump field *)
Pump = Table[{0.0}, {i, 1, Ntot, 1}];
       表格
Pump[[p + 1, 1]] = ϵ / (κnorm * Anorm);
Pump += EA0;


MCa =
  Table[{-ⅈ * Δa[i - 1 - 1/2 × (1 + Sign[i - Nspan - 1.5]) Ntot] - κa} / κnorm, {i, 1, Ntot}];
  表格                          正负符号

MCb = Table[{-ⅈ * Δb[i - 1 - 1/2 × (1 + Sign[i - Nspan - 1.5]) Ntot] - κb} / κnorm, {i, 1, Ntot}];
      表格                          正负符号

MCc = Table[{-ⅈ * Δc[i - 1 - 1/2 × (1 + Sign[i - Nspan - 1.5]) Ntot] - κb} / κnorm, {i, 1, Ntot}];
      表格                          正负符号

MCr =
 Table[{-ⅈ * Δr[i - 1 - 1/2 × (1 + Sign[i - Nspan - 1.5]) Ntot] - κr} / κnorm, {i, 1, Ntot}];
 表格                          正负符号


Idd = IdentityMatrix[Ntot];
      单位矩阵
MCI = Table[{1.0}, {i, 1, Ntot}];
      表格


G2aab = -2 ⅈ * g2aab * √Ntot * Anorm/κnorm * MCI + EA0;
(*why we do the fourier transform here for the interaction parameters?*)
G2abc = -ⅈ * g2abc * √Ntot * Anorm/κnorm * MCI + EA0;
G3aa = -2 ⅈ * g3aa * Ntot * Anorm^2/κnorm * MCI + EA0;
G3ab = -ⅈ * g3ab * Ntot * Anorm^2/κnorm * MCI + EA0;
G3bb = -2 ⅈ * g3bb * Ntot * Anorm^2/κnorm * MCI + EA0;
G3ac = -ⅈ * g3ac * Ntot * Anorm^2/κnorm * MCI + EA0;
GR = -ⅈ * gR * √Ntot * Anorm^2/κnorm * MCI + EA0;
```

```
In[243]:= (* FFT treatment ????*)
Pump = InverseFourier[Fourier[Pump]];
       离散傅立叶逆变换  傅立叶
G2aab = InverseFourier[Fourier[G2aab]];
        离散傅立叶逆变换  傅立叶
G2abc = InverseFourier[Fourier[G2abc]];
        离散傅立叶逆变换  傅立叶
G3aa = InverseFourier[Fourier[G3aa]];
       离散傅立叶逆变换  傅立叶
G3ab = InverseFourier[Fourier[G3ab]];
       离散傅立叶逆变换  傅立叶
G3bb = InverseFourier[Fourier[G3bb]];
       离散傅立叶逆变换  傅立叶
G3ac = InverseFourier[Fourier[G3ac]];
       离散傅立叶逆变换  傅立叶
GR = InverseFourier[Fourier[GR]];
     离散傅立叶逆变换  傅立叶
MCa = InverseFourier[Fourier[MCa]];
      离散傅立叶逆变换  傅立叶
MCb = InverseFourier[Fourier[MCb]];
      离散傅立叶逆变换  傅立叶

In[253]:= (*time scale*)
nanosec = 1.0 * 10^-9;
picosec = 1.0 * 10^-12;
T = 1 nanosec * κnorm;

In[256]:=
(* Start the Simulation *)
lifig = {};
t1 = AbsoluteTime[] - 3 690 574 986;
     绝对时间长度
Tglobal = 0.0;

(*time step*)
dt = 0.008 picosec * κnorm; Ndt = Round[0.01 T / dt];
                                  舍入
PulseLength = 40.;
For[iii = 0, iii ≤ PulseLength * Ndt, iii += Ndt,
For循环

  (*Isb=Abs[A[[1+Nspan* 0+1,1]]]^2/ϵ^2;*)
        绝对值
  IsA = Sum[Abs[EA[[kkkk, 1]]]^2, {kkkk, 1, Ntot}];
        求和
  IsB = Sum[Abs[EB[[kkkk, 1]]]^2, {kkkk, 1, Ntot}];
        求和
  IsC = Sum[Abs[EC[[kkkk, 1]]]^2, {kkkk, 1, Ntot}];
        求和
  IsR = Sum[Abs[ER[[kkkk, 1]]]^2, {kkkk, 1, Ntot}];
        求和
  If[IsA ≥ 1 000 000 000 000, iii = 100 000 000; Break[]];
  如果                                           跳出循环
```

```
AppendTo[liNpA, {Tglobal, IsA * Anorm^2 + 0.01}];
附加
AppendTo[liNpB, {Tglobal, IsB * Anorm^2 + 0.01}];
附加
AppendTo[liNpC, {Tglobal, IsC * Anorm^2 + 0.01}];
附加
AppendTo[liNpR, {Tglobal, IsR * Anorm^2 + 0.01}];
附加


For[jjj = 1, jjj ≤ Ndt, jjj += 1,
For循环

 Tglobal += dt/κnorm * 10^9;

 (*1*)
 Fa = Fourier[EA]; Fb = Fourier[EB]; Fc = Fourier[EC]; Fr = Fourier[ER];
      傅立叶               傅立叶               傅立叶               傅立叶
 TWMa = InverseFourier[Conjugate[Fa] * Fb];
        离散傅立叶逆变换   共轭
 TWMb = InverseFourier[Fa^2];
        离散傅立叶逆变换
 FWMaa = InverseFourier[Fa^2 * Conjugate[Fa]];
         离散傅立叶逆变换       共轭
 FWMab = InverseFourier[Fa * Conjugate[Fb] * Fb];
         离散傅立叶逆变换     共轭
 FWMbb = InverseFourier[Fb^2 * Conjugate[Fb]];
         离散傅立叶逆变换       共轭
 FWMba = InverseFourier[Fa * Conjugate[Fa] * Fb];
         离散傅立叶逆变换     共轭
 k1A = MCa * EA + G2aab * TWMa + G2abc * InverseFourier[Conjugate[Fb] * Fc] +
                                         离散傅立叶逆变换   共轭
   G3aa * FWMaa + G3ab * FWMab + G3ac * InverseFourier[Conjugate[Fa]^2 * Fc] +
                                        离散傅立叶逆变换
   GR * InverseFourier[Fa * Fr] + GR * InverseFourier[Fa * Conjugate[Fr]] + Pump;
        离散傅立叶逆变换                 离散傅立叶逆变换         共轭
 k1B = MCb * EB + G2aab * TWMb + G2abc * InverseFourier[Conjugate[Fa] * Fc] +
                                         离散傅立叶逆变换   共轭
   G3bb * FWMbb + G3ab * FWMba;
 k1C = MCc * EC + G2abc * InverseFourier[Fa * Fb] + G3ac * InverseFourier[Fa^3];
                          离散傅立叶逆变换                  离散傅立叶逆变换
 k1R = MCr * ER + GR * InverseFourier[Conjugate[Fa] * Fa];
                       离散傅立叶逆变换   共轭


 (*2*)
 EAp = EA + dt/2 * k1A;
 EBp = EB + dt/2 * k1B;
```

```
ECp = EC + dt/2 * k1C;
ERp = ER + dt/2 * k1R;
Fa = Fourier[EAp];
     傅立叶
Fb = Fourier[EBp];
     傅立叶
Fc = Fourier[ECp];
     傅立叶
Fr = Fourier[ERp];
     傅立叶
TWMa = InverseFourier[Conjugate[Fa] * Fb];
       离散傅立叶逆变换 共轭
TWMb = InverseFourier[Fa^2];
       离散傅立叶逆变换
FWMaa = InverseFourier[Fa^2 * Conjugate[Fa]];
        离散傅立叶逆变换        共轭
FWMab = InverseFourier[Fa * Conjugate[Fb] * Fb];
        离散傅立叶逆变换      共轭
FWMbb = InverseFourier[Fb^2 * Conjugate[Fb]];
        离散傅立叶逆变换        共轭
FWMba = InverseFourier[Fa * Conjugate[Fa] * Fb];
        离散傅立叶逆变换      共轭
k2A = MCa * EAp + G2aab * TWMa + G2abc * InverseFourier[Conjugate[Fb] * Fc] +
                                         离散傅立叶逆变换 共轭
  G3aa * FWMaa + G3ab * FWMab + G3ac * InverseFourier[Conjugate[Fa]^2 * Fc] +
                                       离散傅立叶逆变换
  GR * InverseFourier[Fa * Fr] + GR * InverseFourier[Fa * Conjugate[Fr]] + Pump;
       离散傅立叶逆变换               离散傅立叶逆变换      共轭
k2B = MCb * EBp + G2aab * TWMb + G2abc * InverseFourier[Conjugate[Fa] * Fc] +
                                         离散傅立叶逆变换 共轭
  G3bb * FWMbb + G3ab * FWMba;
k2C = MCc * ECp + G2abc * InverseFourier[Fa * Fb] + G3ac * InverseFourier[Fa^3];
                          离散傅立叶逆变换                 离散傅立叶逆变换
k2R = MCr * ERp + GR * InverseFourier[Conjugate[Fa] * Fa];
                       离散傅立叶逆变换 共轭


(*3*)
EAp = EA + dt/2 * k2A;
EBp = EB + dt/2 * k2B;
ECp = EC + dt/2 * k2C;
ERp = ER + dt/2 * k2R;
Fa = Fourier[EAp];
     傅立叶
```

```
Fb = Fourier[EBp];
      傅立叶
Fc = Fourier[ECp];
      傅立叶
Fr = Fourier[ERp];
      傅立叶
TWMa = InverseFourier[Conjugate[Fa] * Fb];
        离散傅立叶逆变换   共轭
TWMb = InverseFourier[Fa^2];
        离散傅立叶逆变换
FWMaa = InverseFourier[Fa^2 * Conjugate[Fa]];
         离散傅立叶逆变换    共轭
FWMab = InverseFourier[Fa * Conjugate[Fb] * Fb];
         离散傅立叶逆变换   共轭
FWMbb = InverseFourier[Fb^2 * Conjugate[Fb]];
         离散傅立叶逆变换    共轭
FWMba = InverseFourier[Fa * Conjugate[Fa] * Fb];
         离散傅立叶逆变换   共轭
k3A = MCa * EAp + G2aab * TWMa + G2abc * InverseFourier[Conjugate[Fb] * Fc] +
                                         离散傅立叶逆变换   共轭
   G3aa * FWMaa + G3ab * FWMab + G3ac * InverseFourier[Conjugate[Fa]^2 * Fc] +
                                        离散傅立叶逆变换
   GR * InverseFourier[Fa * Fr] + GR * InverseFourier[Fa * Conjugate[Fr]] + Pump;
        离散傅立叶逆变换              离散傅立叶逆变换     共轭
k3B = MCb * EBp + G2aab * TWMb + G2abc * InverseFourier[Conjugate[Fa] * Fc] +
                                         离散傅立叶逆变换   共轭
   G3bb * FWMbb + G3ab * FWMba;
k3C = MCc * ECp + G2abc * InverseFourier[Fa * Fb] + G3ac * InverseFourier[Fa^3];
                          离散傅立叶逆变换                  离散傅立叶逆变换
k3R = MCr * ERp + GR * InverseFourier[Conjugate[Fa] * Fa];
                       离散傅立叶逆变换   共轭


(*4*)
EAp = EA + dt * k3A;
EBp = EB + dt * k3B;
ECp = EC + dt * k3C;
ERp = ER + dt * k3R;
Fa = Fourier[EAp];
      傅立叶
Fb = Fourier[EBp];
      傅立叶
Fc = Fourier[ECp];
      傅立叶
Fr = Fourier[ERp];
      傅立叶
TWMa = InverseFourier[Conjugate[Fa] * Fb];
        离散傅立叶逆变换   共轭
TWMb = InverseFourier[Fa^2];
        离散傅立叶逆变换
FWMaa = InverseFourier[Fa^2 * Conjugate[Fa]];
         离散傅立叶逆变换    共轭
```

```
FWMab = InverseFourier[Fa * Conjugate[Fb] * Fb];
        └离散傅立叶逆变换      └共轭
FWMbb = InverseFourier[Fb^2 * Conjugate[Fb]];
        └离散傅立叶逆变换        └共轭
FWMba = InverseFourier[Fa * Conjugate[Fa] * Fb];
        └离散傅立叶逆变换      └共轭
k4A = MCa * EAp + G2aab * TWMa + G2abc * InverseFourier[Conjugate[Fb] * Fc] +
                                         └离散傅立叶逆变换 └共轭
  G3aa * FWMaa + G3ab * FWMab + G3ac * InverseFourier[Conjugate[Fa]^2 * Fc] +
                                       └离散傅立叶逆变换
  GR * InverseFourier[Fa * Fr] + GR * InverseFourier[Fa * Conjugate[Fr]] + Pump;
       └离散傅立叶逆变换               └离散傅立叶逆变换      └共轭
k4B = MCb * EBp + G2aab * TWMb + G2abc * InverseFourier[Conjugate[Fa] * Fc] +
                                         └离散傅立叶逆变换 └共轭
  G3bb * FWMbb + G3ab * FWMba;
k4C = MCc * ECp + G2abc * InverseFourier[Fa * Fb] + G3ac * InverseFourier[Fa^3];
                          └离散傅立叶逆变换               └离散傅立叶逆变换
k4R = MCr * ERp + GR * InverseFourier[Conjugate[Fa] * Fa];
                       └离散傅立叶逆变换 └共轭

 (*end*)
EA += 1/6 * dt * (k1A + 2 k2A + 2 k3A + k4A);
EB += 1/6 * dt * (k1B + 2 k2B + 2 k3B + k4B);
EC += 1/6 * dt * (k1C + 2 k2C + 2 k3C + k4C);
ER += 1/6 * dt * (k1R + 2 k2R + 2 k3R + k4R)
];
lica = Abs[Transpose[Fa]]^2;
licb = Abs[Transpose[Fb]]^2;
licc = Abs[Transpose[Fc]]^2;

CombAav += EA * Conjugate[EA];
                └共轭
CombBav += EB * Conjugate[EB];
                └共轭
CombCav += EC * Conjugate[EC];
                └共轭

CombA = Table[{(2 π * 3. * 10^8)/(liωa[[i, 1]]) * 10^9,
        └表格
  10 * Log[10, hbar * ω0a * 2 κa1 * Abs[EA[[i, 1]]]^2 * Abs[Anorm^2] + 10^-199]}, {i, 1, Ntot}];
       └对数                                          └绝对值

CombB = Table[{(2 π * 3. * 10^8)/(liωb[[i, 1]]) * 10^9, 10 * Log[10,
        └表格                                            └对数
```

```
        hbar * ω0b * 2 κb1 * Abs[EB[[i, 1]]]^2 * Abs[Anorm^2] + 10^-199]}, {i, 1, Ntot}];
                                                  绝对值

    CombC = Table[{ (2 π * 3. * 10^8) / (liωc[[i, 1]]) * 10^9, 10 * Log[10,
              表格                                                   对数

        hbar * ω0c * 2 κb1 * Abs[EC[[i, 1]]]^2 * Abs[Anorm^2] + 10^-199]}, {i, 1, Ntot}]
                                                  绝对值
    ];

t2 = AbsoluteTime[] - 3 690 574 986;
       绝对时间长度
Print["Time used: ", t2 - t1, " Seconds"];
打印
```

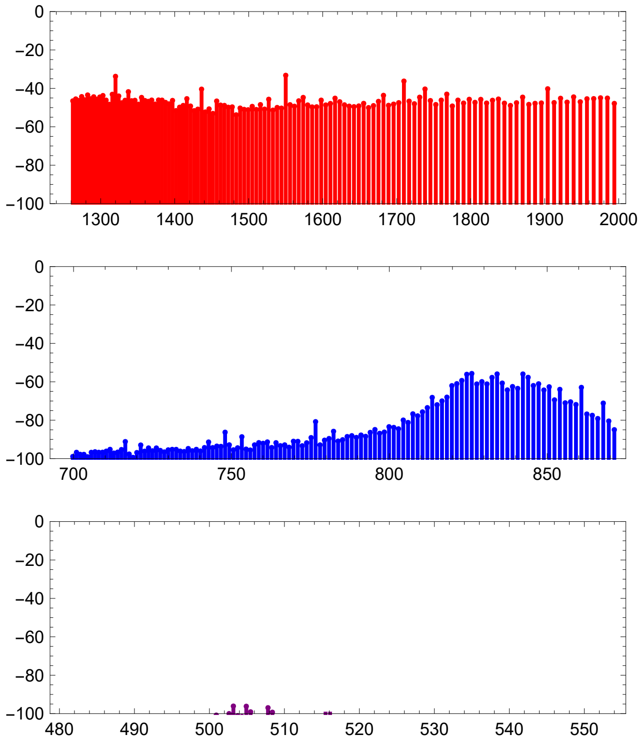
Time used: 184.2118773 Seconds

```
In[264]:= PBtot = Sum[hbar * ω0b * 2 κb1 * Abs[Anorm]^2 * CombBav[[i, 1]] / PulseLength, {i, 1, Ntot}];
                 求和
PCtot = Sum[hbar * ω0c * 2 κb1 * Abs[Anorm]^2 * CombCav[[i, 1]] / PulseLength, {i, 1, Ntot}];
        求和
PAtot = Sum[hbar * ω0a * 2 κa1 * Abs[Anorm]^2 * CombAav[[i, 1]] / PulseLength, {i, 1, Nspan}] +
        求和
   Sum[hbar * ω0a * 2 κa1 * Abs[Anorm]^2 * CombAav[[i, 1]] / PulseLength, {i, 2 + Nspan, Ntot}];
   求和
Print["The power of A,B,C are ", Re[PAtot], "Watt, ",
打印                              常量  实部
  Re[PBtot], "Watt", Re[PCtot], "Watt"];
  实部                实部
```

The power of A,B,C are 0.00374447Watt, 0.0000253316Watt1.15326 × 10^{-9}Watt

```
In[268]:= CombAavout = Table[{(2 π * 3. * 10^8)/(liωa⟦i, 1⟧) * 10^9,
                   表格
      10 * Log[10, hbar * ω0a * 2 κa1 * Abs[Anorm^2] * CombAav⟦i, 1⟧ / PulseLength + 10^-199]}, {i,
           对数                           绝对值
      1, Ntot}];
   CombBavout = Table[{(2 π * 3. * 10^8)/(liωb⟦i, 1⟧) * 10^9,
                表格
      10 * Log[10, hbar * ω0b * 2 κb1 * Abs[Anorm^2] * CombBav⟦i, 1⟧ / PulseLength + 10^-199]}, {i,
           对数                           绝对值
      1, Ntot}];
   CombCavout = Table[{(2 π * 3. * 10^8)/(liωc⟦i, 1⟧) * 10^9,
                表格
      10 * Log[10, hbar * ω0c * 2 κc1 * Abs[Anorm^2] * CombCav⟦i, 1⟧ / PulseLength + 10^-199]}, {i,
           对数                           绝对值
      1, Ntot}];
   figav = GraphicsGrid[{{ListPlot[CombAavout, PlotStyle → Red, Filling → -200,
           图形网格        绘制点集                绘图样式     红色   填补
        FillingStyle → Directive[Thick, Red], ImageSize → 400, AspectRatio → 1 / 3,
        填充样式         指令       粗     红色   图像尺寸           宽高比
        PlotRange → {All, {-100, 0}}, Frame → True, Axes → False]},
        绘制范围     全部                 边框    真     坐标轴  假
      {ListPlot[CombBavout, PlotStyle → Blue, Filling → -200,
       绘制点集                绘图样式     蓝色   填补
        FillingStyle → Directive[Thick, Blue], ImageSize → 400, AspectRatio → 1 / 3,
        填充样式         指令       粗     蓝色    图像尺寸           宽高比
        PlotRange → {All, {-100, 0}}, Frame → True, Axes → False]},
        绘制范围     全部                 边框    真     坐标轴  假
      {ListPlot[CombCavout, PlotStyle → Purple, Filling → -200,
       绘制点集                绘图样式     紫色     填补
        FillingStyle → Directive[Thick, Purple], ImageSize → 400, AspectRatio → 1 / 3,
        填充样式         指令       粗     紫色      图像尺寸           宽高比
        PlotRange → {All, {-100, 0}}, Frame → True, Axes → False]}}];
        绘制范围     全部                 边框    真     坐标轴  假
```

In[272]:=
```
Print[figav];
打印
```



In[273]:=
```
(*rig=ToString[Round[Random[]*1000000]];*)
  转换为…  舍入
rig = ToString[kkkk];
  转换为字符串
Mname = $MachineName;
  机器名称
lidate = DateList[];
  日期列表
timeStr = ToString[lidate[[1]]] <> ToString[lidate[[2]]] <>
  转换为字符串  转换为字符串
   ToString[lidate[[3]]] <> ToString[lidate[[4]]] <> ToString[lidate[[5]]];
   转换为字符串  转换为字符串  转换为字符串
nb = InputNotebook[];
  输入笔记本
NotebookPrint[nb, dir <> rig <> "-" <> Mname <> "-" <> timeStr <> ".pdf"];
笔记本打印
Export[rig <> "-" <> Mname <> "-" <> timeStr <> "-average.png", figav]
导出
```

Out[•]= kkkk-laptop-fi64hr6k-202411122113-average.png